

Generalized Linear Models

EES 4891/5891

Probability & Statistics for Geosciences

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Class #22: Tuesday, April 01 2025

Getting Started

Getting Started

- Accept the GitHub Classroom assignment at <https://classroom.github.com/a/WJDv3nRt>



- Note: A lot of the examples and exercises in today's class come from the textbook, *Hurricane Climatology* by James B. Elsner and Thomas H. Jagger (Oxford University Press, 2013).

Learning Goals

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- Learn about how to use regression analysis with discrete variables
- Understand what Generalized Linear Modeling is and how it differs from Linear Regression Modeling
- Understand the basics of logistic regression
 - Understand how to do logistic regression
- Understand how to assess logistic regression models
 - Know what an ROC curve is, how to plot one in R, and how to interpret one
- Learn other Generalized Linear Models: Poisson Regression
- What is *forecast skill* and how would you use it to assess a model?

Linear Regression with Discrete Variables

Linear Regression with Discrete Variables

- Continuous variables:
 - Temperature, mass, pH, silica content, grain size, ...
- Discrete variables:
 - Count data: How many earthquakes, volcanic eruptions, hurricanes, ...
 - Categorical data: Mineral type (quartz, calcite, biotite, ...), species, fault type, ...
- Four possibilities:
 - Continuous independent, continuous dependent
 - Continuous independent, discrete dependent
 - Discrete independent, continuous independent
 - Discrete independent, discrete dependent
- Categorical vs. Ordinal discrete variables:
 - Ordinal: count, ordered categories (age-range, etc.)
 - One measurement can be greater or less than another
 - This doesn't make sense with unordered categories:
 - Mineral type, species, ...

Discrete Variables

- Ordinal independent variables:
 - Count,
 - Mostly, treat them like continuous variables
- Categorical independent variables:
 - “One-hot” encoding (dummy variables)
 - If x has k possible values, replace x with k different variables,
 - One and only one is 1, and the rest are 0
 - Instead of one column, “mineral,” have “quartz”, “calcite”, “K-feldspar”, “plagioclase feldspar”, “mica”, “amphibole”, ...
- Ordinal dependent variables:
 - Count data:
 - $x \sim \mathcal{D}$, a discrete distribution (Bernoulli, Binomial, Poisson, etc.)
 - Parameters:
 - Poisson has a continuous parameter $\lambda \geq 0$.
 - Bernoulli, Binomial, etc. have a continuous parameter probability $p : 0 \leq p \leq 1$.
 - Logistic regression
- Categorical dependent variables:
 - Multinomial distributions.
 - We won’t cover this.

Logistic Regression

- Linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- Count data:

$$y \sim \text{Binomial}(p(x), n)$$

$$\mathbb{P}(y = k | p(x), n) = \binom{n}{k} p(x)^k (1 - p(x))^{n-k}$$

$$p(x) \sim \mathcal{D}(\theta, x)$$

We're used to regression of the form

$$p(x) = \beta_0 + \beta_1 x + \varepsilon$$

but how do we constrain p to be between 0 and 1?

- Logit function:

$$\text{logit}(p) = \ln \frac{p}{1-p}$$

$$\text{logit}^{-1}(x) = \frac{e^x}{e^x + 1}$$

- Logit maps the interval $(0, 1)$ to the interval $(-\infty, \infty)$
- Inverse-logit maps the interval $(-\infty, \infty)$ to the interval $(0, 1)$

- Logistic regression:

$$y \sim \text{Binomial}(x, n)$$

$$z = \beta_0 + \beta_1 x + \varepsilon$$

$$p = \text{logit}^{-1}(z)$$

The Logit Function

- Logit function:

$$\text{logit}(p) = \ln \frac{p}{1-p}$$

$$\text{logit}^{-1}(x) = \frac{e^x}{e^x + 1}$$

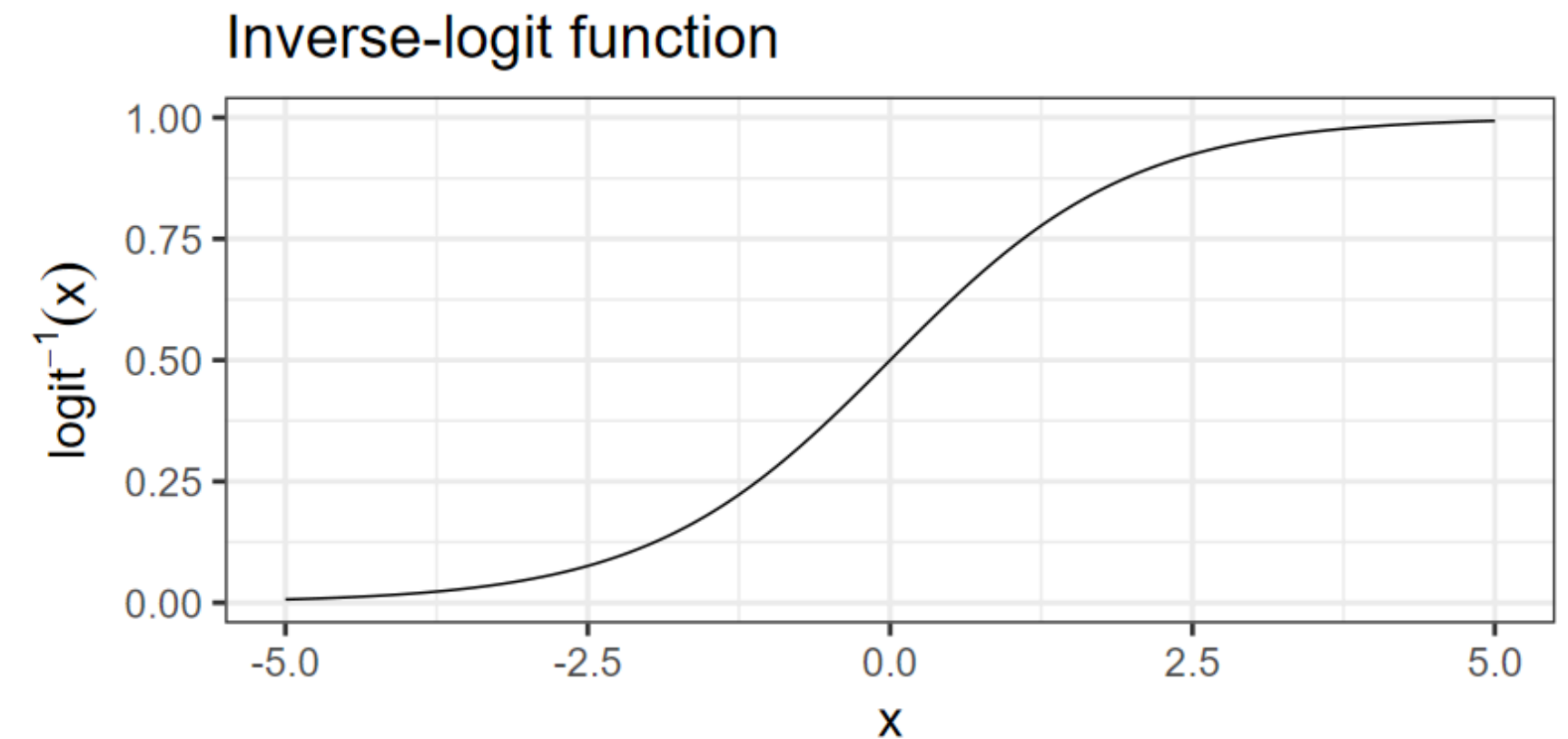
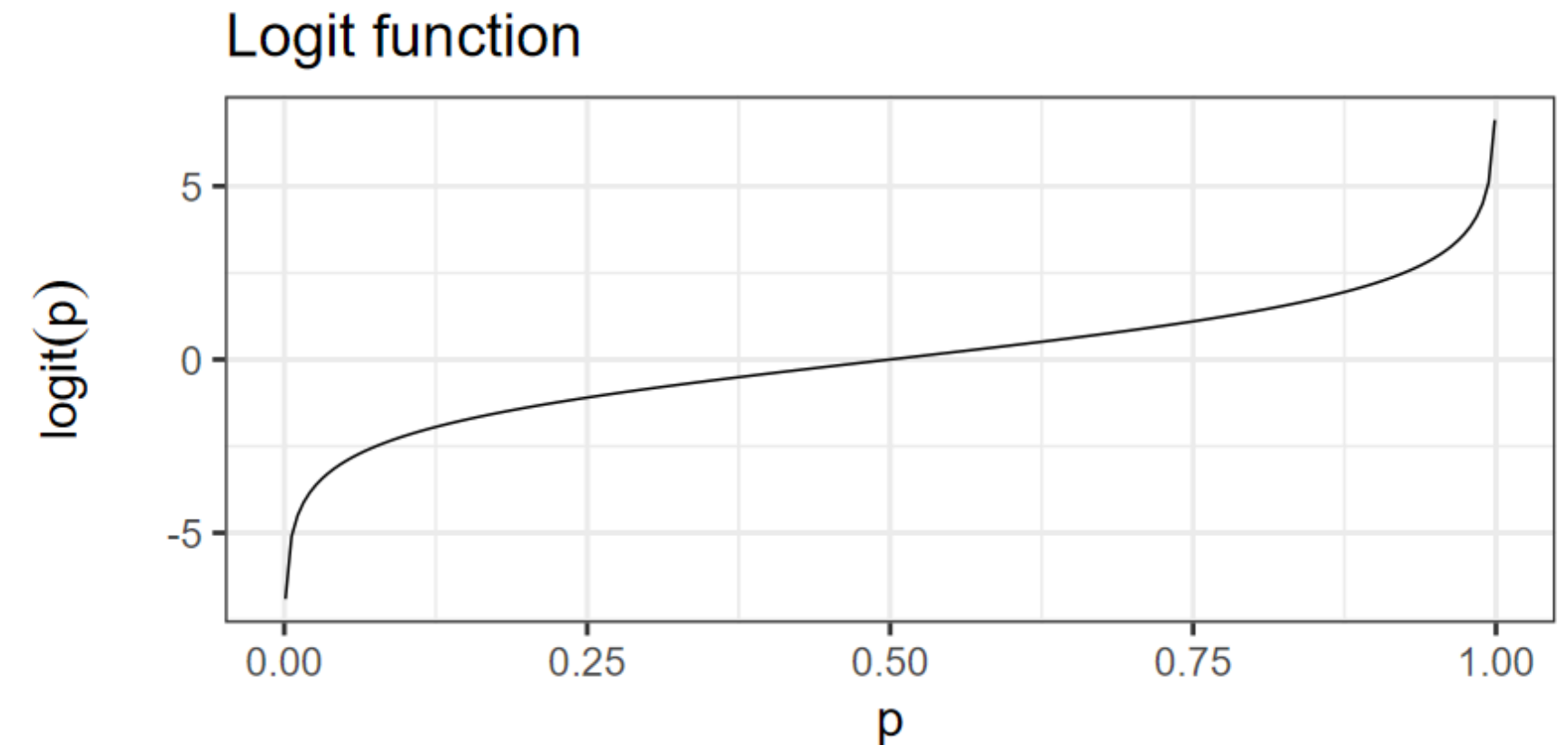
- Logistic regression:

$$y \sim \text{Binomial}(x, n)$$

$$z = \beta_0 + \beta_1 x + \varepsilon$$

$$p = \text{logit}^{-1}(z)$$

- z ranges anywhere between $-\infty$ and ∞ .
- The $\text{logit}^{-1}(z)$ function maps it onto the $(0, 1)$ interval suitable for a probability.



Generalized Linear Models

Generalized Linear Models

- Linear Regression Model:

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j} + \varepsilon_i$$

$$E(Y|X) = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j}$$

- Generalized Linear Model:

- Three parts:

1. A distribution $\mathcal{D}$ for Y

- Bernoulli (coin-toss), Binomial, Poisson, etc.

2. A linear predictor $\eta_i = \beta_0 + \sum_j \beta_j x_{i,j}$

3. A link function $g(\eta)$

- logit, logarithm, etc.

$$y_i \sim \mathcal{D} (\mu = g^{-1}(\eta_i))$$

$$\eta_i = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j} + \varepsilon_i$$

$$E(y_i|x_{i,j}) = g^{-1} \left(\beta_0 + \sum_{j=1}^p \beta_j x_{i,j} \right)$$

Link Functions & Probability Distributions

- Three parts:
 1. A distribution $\mathcal{D}$ for Y
 - Bernoulli (coin-toss), Binomial, Poisson, etc.
 2. A linear predictor $\eta_i = \beta_0 + \sum_j \beta_j x_{i,j}$
 3. A link function $g(\eta)$
 - logit, logarithm, etc.
- The parameters θ of $\mathcal{D}(\theta)$ should be related to g so $\mu = g^{-1}(\eta_i)$ is the mean for the probability distribution $\mathcal{D}(\theta)$.
 - This means $\mu = g^{-1}(\eta_i)$ is the expectation value for y_i

- Examples:

Distribution	Values	Link	Link function
Normal	$(-\infty, \infty)$	Identity	$g(\mu) = \mu$
Gamma	$[0, \infty)$	Negative inverse	$g(\mu) = -1/\mu$
Poisson	0, 1, 2, ...	Log	$g(\mu) = \ln(\mu)$
Bernoulli	0, 1	Logit	$g(\mu) = \ln\left(\frac{\mu}{1-\mu}\right)$
Binomial	0, 1, ..., N	Logit	$g(\mu) = \ln\left(\frac{\mu}{N-\mu}\right)$

Logistic Regression in R

Hurricane Climatology

- Cyclonic storm systems can be categorized as
 - **Tropical** (warm-core, *barotropic*).
 - **Extratropical** (cold-core, *baroclinic*).
 - *barotropic* systems don't exhibit wind shear, *baroclinic* systems do.
- Hurricanes are barotropic tropical cyclones, but they can have baroclinic influences.
- Challenge: Use logistic regression to predict which hurricanes have baroclinic influences.
- Theoretical basis:
 - Most hurricanes form in the deep tropics, over uniformly warm seas: barotropic, no wind shear.
 - Some hurricanes form at higher latitudes, where interaction with midlatitude jet streams affect them (baroclinic).
- Use logistic regression against the latitude at which the hurricane formed to predict baroclinic influence.

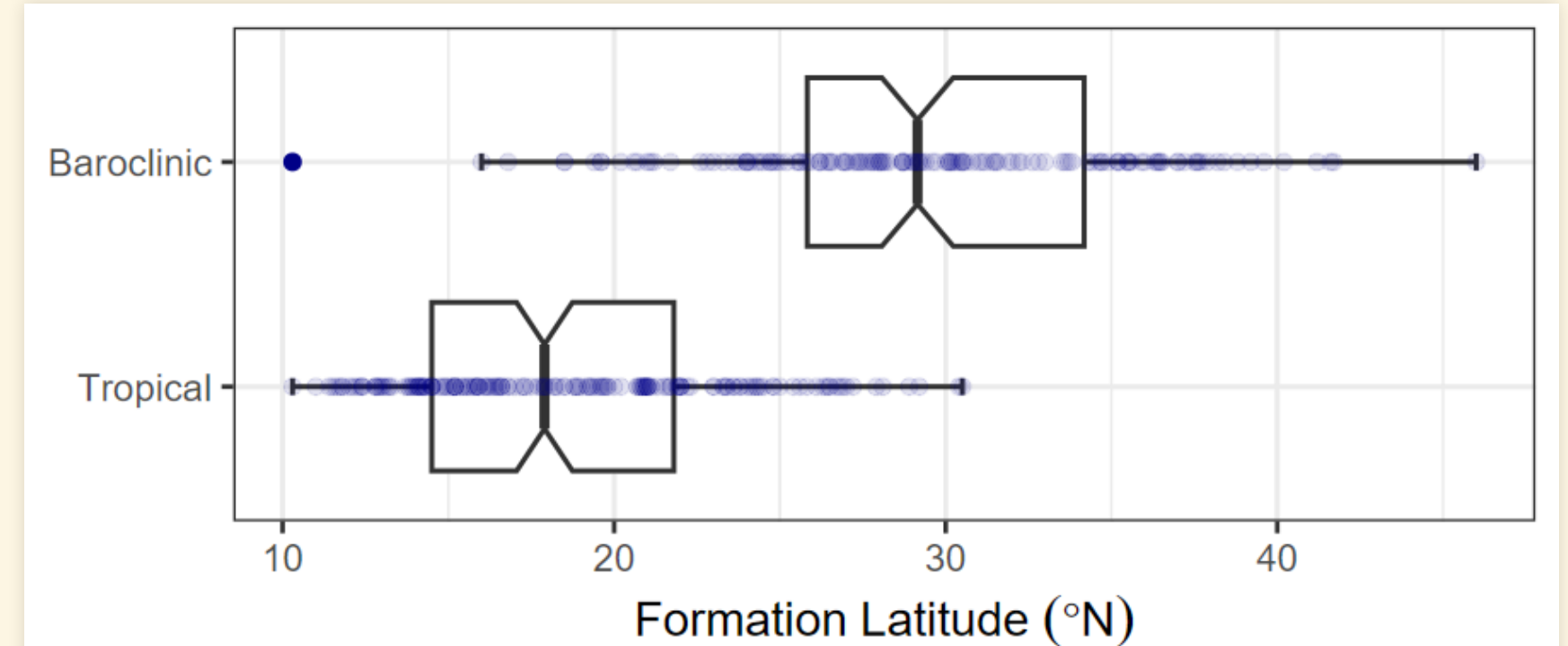
Hurricane Data

- Load data:
 - A data set for 337 hurricanes. The column `Type` is a factor indicating whether the cyclone is tropical or baroclinic. `FirstLat` gives the latitude at which the hurricane formed.

```
bh <- read_rds("bh.rds")
glimpse(bh)
```

```
## Rows: 337
## Columns: 10
## $ Name      <chr> "NOTNAMED", "NOTNAMED", "NOTNAM...
## $ Year      <dbl> 1944, 1944, 1944, 1944, 1944, 1...
## $ Type      <fct> Baroclinic, Tropical, Tropical,...
## $ FirstLat  <dbl> 30.2, 25.6, 14.2, 20.8, 20.0, 2...
## $ FirstLon  <dbl> -76.1, -74.9, -65.2, -58.0, -84...
## $ MaxLat    <dbl> 32.1, 31.0, 16.6, 26.3, 20.6, 3...
## $ MaxLon    <dbl> -74.8, -78.1, -72.2, -72.3, -84...
## $ LastLat   <dbl> 35.1, 32.6, 20.6, 42.1, 19.1, 5...
## $ LastLon   <dbl> -69.2, -78.2, -88.5, -71.5, -93...
## $ MaxInt    <dbl> 80, 80, 105, 120, 70, 85, 105, ...
```

- Begin by plotting data

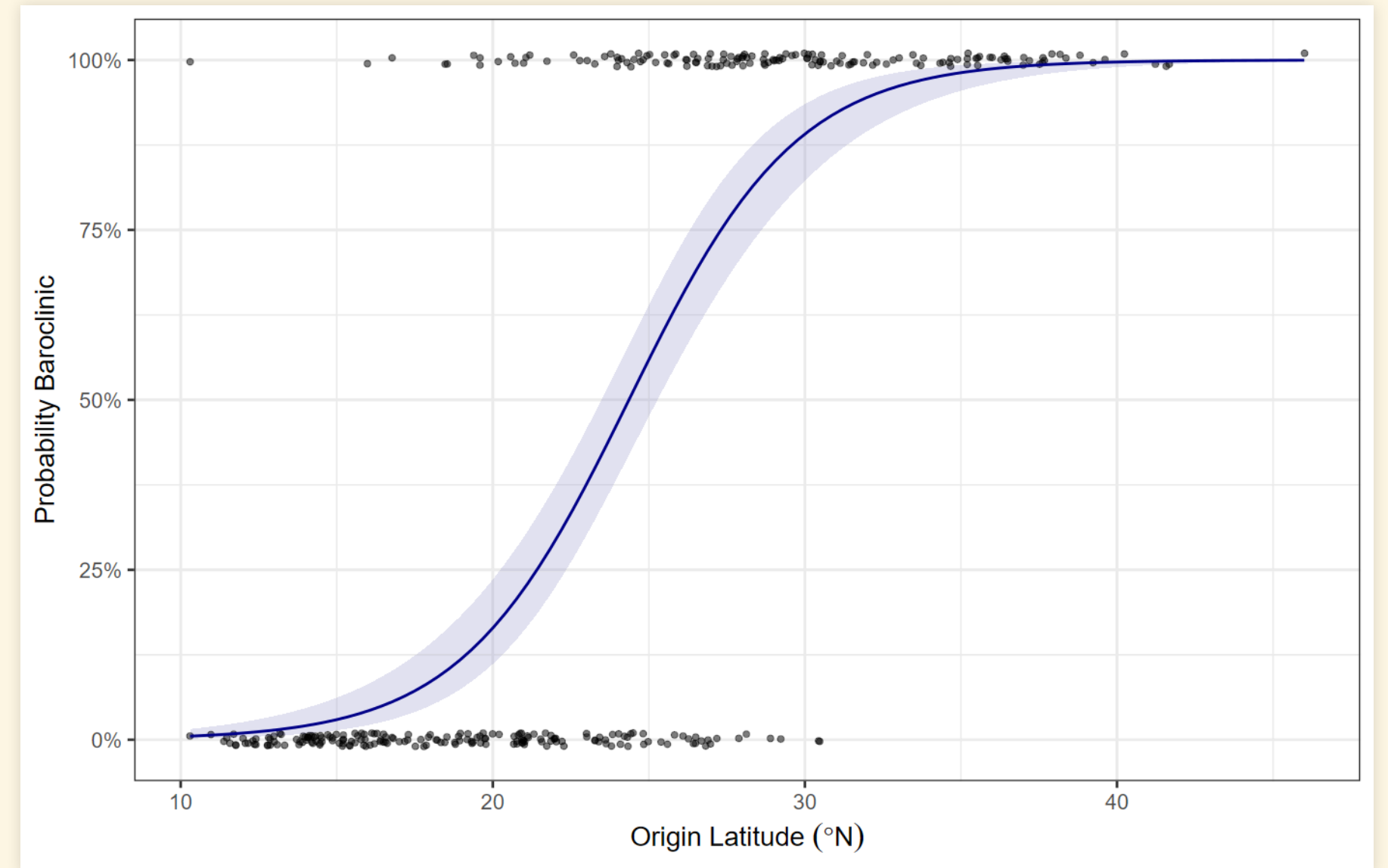


Logistic Regression Model

- Make the model, using logistic regression with R's "glm" engine for fitting the parameters:

```
recipe <- recipe(bh, Type ~ FirstLat)
model <- logistic_reg() |>
  set_engine("glm", family = "binomial")
wflow <- workflow(recipe) |> add_model(model)
results <- wflow |> fit(data = bh)
tidy(results)
```

```
## # A tibble: 2 × 5
##   term      estimate std.error statistic  p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept) -9.08      0.961     -9.45 3.50e-21
## 2 FirstLat    0.373     0.0395      9.45 3.49e-21
```



Assessing the Quality of a Logistic Model

- We measure linear regression models by the sum of the squares of the residuals (errors)
- But a logistic model predicts a probability of something that's 0 or 1. How do you measure the error?
- One way: ROC (Receiver Operating Characteristic) curves.
 1. Split your data into *training* and *test* sets.
 - You can also use *k*-fold cross-validation.
 2. Fit your model to the *training* set.
 3. Generate predictions for the *test* set.
 - Now you have a list of predicted *probabilities* and observed categories
 4. Vary a *threshold* x : Predict that a hurricane is baroclinic if $p_{\text{baroclinic}} > x$.
 - For each value of x , count false positives and false negatives.
- ROC Curve
 5. Plot *sensitivity* vs $1 - \text{specificity}$.
 - *sensitivity*: Probability of correct test if H_0 is false.
 - True positive if “positive” is the correct answer.
 - Probability of correctly identifying Baroclinic storms.
 - *specificity*: Probability of correct test if H_0 is true.
 - True negative if “negative” is the correct answer.
 - Probability of correctly identifying Tropical storms.

ROC Curves in R

```
set.seed(12345)
# strata=Type ensures that each fold has a good balance of
# Tropical and Baroclinic cyclones.
k_fold <- vfold_cv(bh, 5, strata = Type)
control <- control_resamples(save_pred = TRUE)

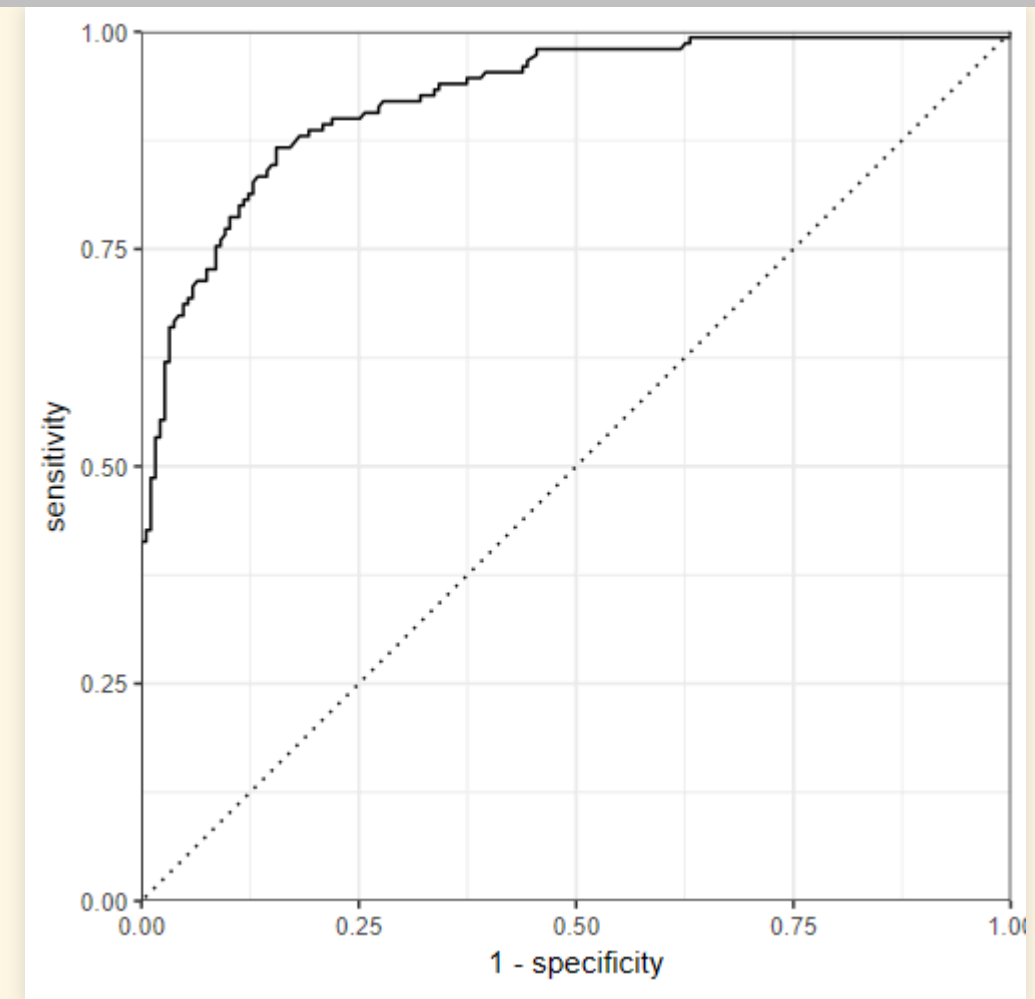
results <- fit_resamples(wflow, k_fold, control = control)
results |> collect_predictions() |> glimpse()
```

```
## Rows: 337
## Columns: 7
## $ .pred_class      <fct> Baroclinic, Tropical, T...
## $ .pred_Tropical   <dbl> 0.1500488, 0.8752090, 0...
## $ .pred_Baroclinic <dbl> 0.84995117, 0.12479104,...
## $ id              <chr> "Fold1", "Fold1", "Fold..."
## $ .row            <int> 6, 10, 15, 16, 17, 19, ...
## $ Type            <fct> Baroclinic, Tropical, T...
## $ .config          <chr> "Preprocessor1_Model1",...
```

```
results |> collect_metrics() |> filter(.metric ==
  "roc_auc")
```

```
## # A tibble: 1 × 6
##   .metric .estimator mean      n std_err .config
##   <chr>   <chr>     <dbl> <int>   <dbl> <chr>
## 1 roc_auc binary    0.927     5 0.0148 Preprocessor1_Model1
```

```
results |> collect_predictions() |>
  roc_curve(truth = Type, .pred_Baroclinic,
            event_level = "second") |>
  autoplot() +
  scale_x_continuous(expand = c(0,0)) +
  scale_y_continuous(expand = c(0,0))
```



- The area under the curve (AUC) measures how accurate the predictions are.
 - 1 = perfect
 - 0.5 = like tossing a coin
 - 0 = perversely gets every prediction wrong

Poisson Regression

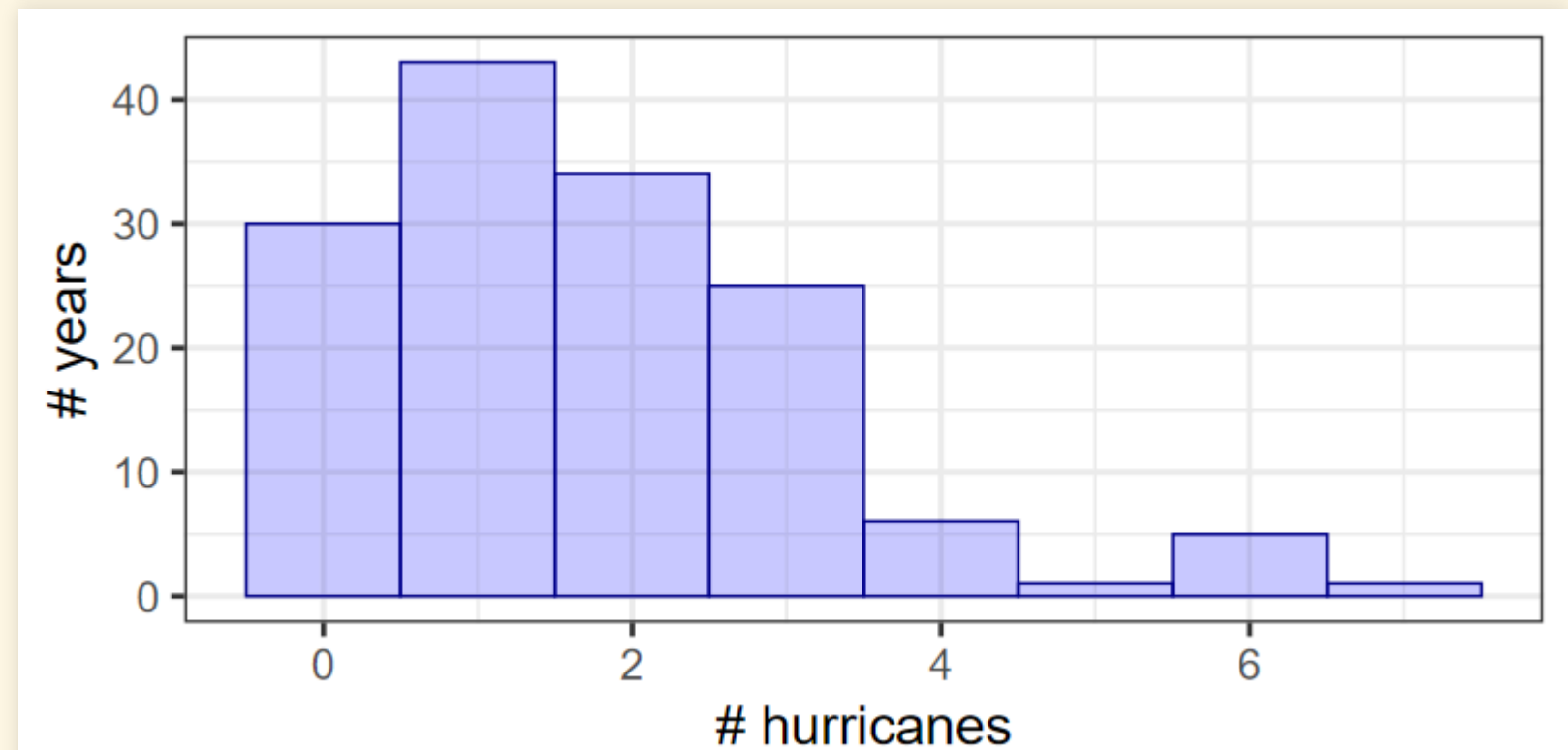
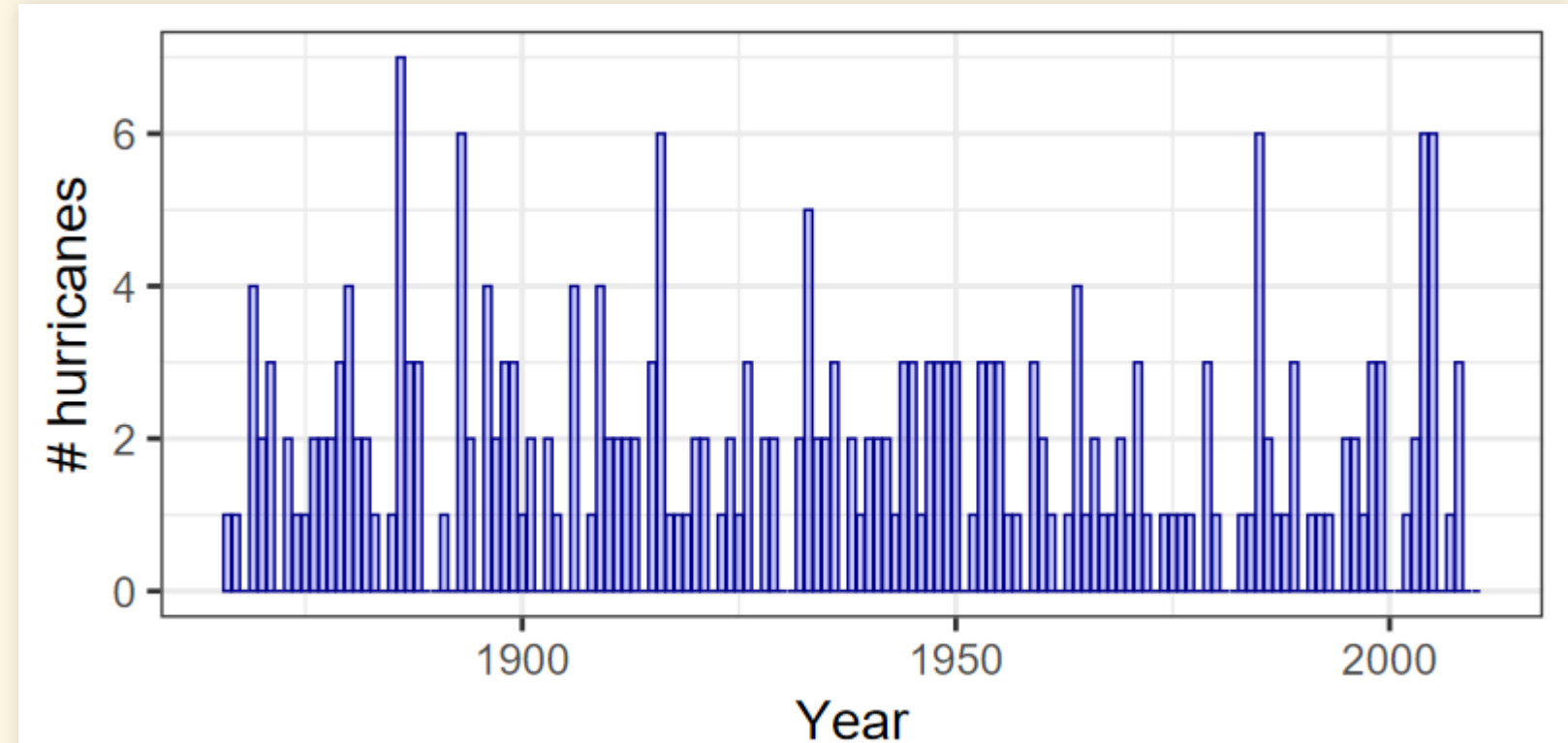
Hurricane Landfalls on US

- Here, we are going to study climatic influences on the number of hurricanes that make landfall in the US
- First, load the data, with the number of hurricanes each year from 1866 to 2010

```
hurricanes <- read_rds("hurricanes.rds")  
glimpse(hurricanes)
```

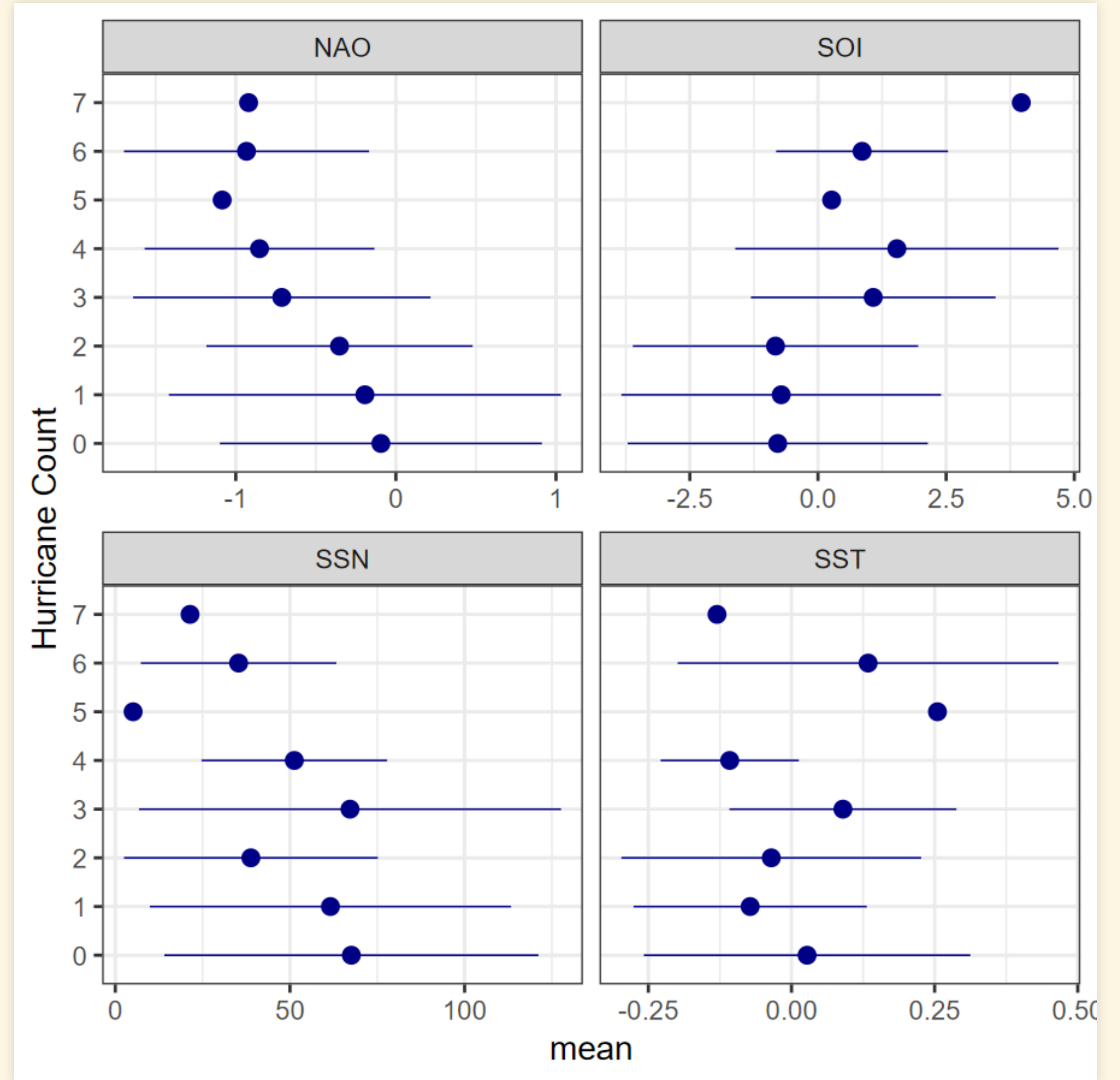
```
## Rows: 145  
## Columns: 10  
## $ Year <dbl> 1866, 1867, 1868, 1869, 1870, 1871,...  
## $ All <dbl> 1, 1, 0, 4, 2, 3, 0, 2, 1, 1, 2, 2,...  
## $ MUS <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1,...  
## $ G <dbl> 1, 1, 0, 2, 1, 0, 0, 0, 0, 1, 0, 1,...  
## $ FL <dbl> 0, 0, 0, 0, 1, 2, 0, 2, 1, 0, 1, 2,...  
## $ E <dbl> 0, 0, 0, 2, 0, 1, 0, 0, 1, 0, 1, 0,...  
## $ NAO <dbl> -0.860, -1.775, 1.725, -1.345, 1.05...  
## $ SOI <dbl> 0.2666667, -0.1333333, -4.0000000, ...  
## $ SST <dbl> -0.02873733, -0.19373733, -0.126737...  
## $ SSN <dbl> 7.3, 9.8, 47.2, 80.6, 136.0, 80.3, ...
```

- Plot hurricane frequency:



Modeling Hurricanes

- Hurricane landfalls clearly follow a Poisson distribution. The average rate of landfalls is 1.737931.
- However, we don't think the number is purely random: climate factors are likely to influence the number of hurricanes.
 - SST: Average Sea-Surface Temperature in the Atlantic Ocean.
 - NAO: North-Atlantic Oscillation Index.
 - SOI: Southern Oscillation Index (related to El Niño).
 - SSN: Sun-spot number, which correlates with the brightness of the sun.



Poisson Regression


```
library(poissonreg)
rec <- recipe(hurricanes, All ~ SST + NAO + SOI + SSN)
mdl <- poisson_reg(engine = "glm")
wflow <- workflow(rec) |> add_model(mdl)

fit <- fit(wflow, hurricanes)

tidy(fit)
```

```
## # A tibble: 5 × 5
##   term          estimate std.error statistic
##   <chr>          <dbl>    <dbl>    <dbl>
## 1 (Intercept)  0.595      0.103      5.76
## 2 SST          0.229      0.255      0.897 0.370
## 3 NAO         -0.167      0.0644     -2.59 0.00972
## 4 SOI          0.0619      0.0213      2.90 0.00371
## 5 SSN         -0.00231     0.00137     -1.68 0.0928
```

- SST doesn't make a significant contribution, so make a new model without SST

```
rec_new <- recipe(hurricanes, All ~ NAO + SOI + SSN)
wflow_new <- workflow(rec_new) |> add_model(mdl)
fit_new <- fit(wflow_new, hurricanes)
```

- New model fit:

```
tidy(fit_new)

## # A tibble: 4 × 5
##   term          estimate std.error statistic
##   <chr>          <dbl>    <dbl>    <dbl>
## 1 (Intercept)  0.585      0.103      5.69
## 2 NAO         -0.177      0.0633     -2.80 0.00507
## 3 SOI          0.0615      0.0213      2.89 0.00387
## 4 SSN         -0.00220     0.00137     -1.60 0.109
```

- Test model performance with cross-validation

```
k_fold <- vfold_cv(hurricanes, 5)
control <- control_resamples(save_pred = TRUE)
results <- fit_resamples(wflow_new, k_fold,
                        control = control)

results |> collect_metrics()
```

```
## # A tibble: 2 × 6
##   .metric .estimator mean      n std_err
##   <chr>   <chr>    <dbl> <int>  <dbl> <chr>
## 1 rmse    standard    1.46     5  0.122
## 2 rsq      standard    0.128     5  0.0602
```


Forecast Skill

- How do we know how good a forecasting model is?
- Skill: how does the forecast model compare to a naïve model?
 - Naïve model: Every year has the same forecast: the average number of hurricanes 1.7
 - RMSE for the naive model is

```
MSE_naive <- hurricanes |>
  summarize(MSE = sum( (All - mean(All))^2 ) / (n() -
                    1)) |>
  pull(MSE)
```

```
MSE_naive
```

```
## [1] 2.194732
```

- MSE for our predictive model is

```
MSE_skilled <- collect_metrics(results) |>
  filter(.metric == "rmse") |>
  mutate(MSE = mean^2) |> pull(MSE)
MSE_skilled
```

```
## [1] 2.119829
```

- So, MSE for the naïve model is 2.19 and MSE for the predictive model is 2.12, which is 3.4% better.

